

GRAND RAPIDS, MB, Canada

WMO: 718580

Lat: 53.183N

Lon: 99.267W

Elev: 223

StdP: 98.68

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.3	-32.7	-40.6	0.1	-33.9	-38.5	0.1	-32.3	7.6	-16.2	6.8	-17.3	1.4	300	0.651

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	28.6	19.4	26.7	18.3	25.0	17.5	20.8	26.1	19.7	24.6	18.7	23.2	3.7	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	14.2	23.1	17.9	13.2	22.1	16.9	12.4	21.1	61.4	26.2	57.4	24.6	54.0	23.2	26.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.4	6.5	5.8	-38.9	31.8	2.6	1.6	-40.8	32.9	-42.3	33.8	-43.8	34.7	-45.6	35.8		
(5)			-38.4	22.7	2.2	1.6	-40.0	23.9	-41.3	24.8	-42.5	25.7	-44.1	26.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.0	-19.1	-17.0	-9.7	-0.6	6.4	13.6	17.5	16.1	10.7	3.1	-6.9	-15.7		
	DBStd	14.16	8.21	7.67	8.17	6.31	4.75	4.41	3.28	3.52	4.57	4.46	6.77	8.12		
	HDD10.0	4302	902	755	611	321	132	14	0	1	45	218	506	797		
	HDD18.3	6765	1160	988	869	569	369	152	56	86	231	472	756	1056		
	CDD10.0	641	0	0	0	2	21	123	232	192	67	4	0	0		
	CDD18.3	63	0	0	0	0	1	11	30	18	3	0	0	0		
	CDH23.3	771	0	0	0	0	26	167	329	207	42	1	0	0		
	CDH26.7	175	0	0	0	0	4	51	70	44	6	0	0	0		
	WSAvg	2.7	2.5	2.6	2.7	2.7	2.8	2.6	2.6	2.6	2.8	2.9	2.7	2.4		
	PrecAvg	485	15	11	23	26	41	77	79	71	64	45	26	18		
(15) Precipitation	PrecMax	702	36	23	52	78	108	197	192	140	125	98	63	34		
	PrecMin	370	4	3	4	2	2	8	17	14	19	4	3	6		
	PrecStd	86	9	6	13	18	26	42	46	39	26	27	15	9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.7	3.9	11.1	19.8	26.4	32.7	31.1	30.4	27.3	19.2	9.7	3.1		
		MCWB	1.0	1.4	5.3	10.1	14.2	21.9	21.0	20.7	18.8	12.9	5.9	0.8		
	2%	DB	-0.8	0.7	7.9	15.2	22.6	27.5	28.6	27.8	24.2	15.3	6.3	-0.3		
		MCWB	-1.9	-0.8	3.7	8.1	12.7	18.2	19.5	19.6	16.7	10.4	3.7	-1.6		
	5%	DB	-3.6	-1.6	4.7	11.7	19.4	24.8	26.8	25.6	21.5	12.7	4.0	-2.2		
		MCWB	-4.4	-2.9	1.6	5.8	11.0	16.4	18.6	18.5	15.6	9.0	2.2	-3.1		
	10%	DB	-6.1	-4.5	1.9	8.6	15.9	22.5	25.1	23.7	19.0	10.5	1.7	-4.3		
		MCWB	-6.7	-5.6	0.1	4.2	9.5	15.6	18.0	17.6	14.3	7.8	0.5	-5.1		
	0.4%	WB	1.1	1.6	5.7	10.8	15.9	21.5	23.1	21.8	19.5	13.5	6.2	0.7		
		MCDB	2.3	3.2	10.4	18.4	23.9	28.3	27.9	27.8	25.1	17.4	8.9	2.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-2.0	-0.5	3.6	8.2	13.4	19.0	21.0	20.6	17.5	11.2	3.9	-1.4		
		MCDB	-0.9	0.7	7.5	14.4	20.8	24.6	25.9	25.9	22.4	14.4	5.9	-0.4		
	5%	WB	-4.4	-2.9	1.7	6.3	11.6	17.6	19.9	19.5	16.1	9.4	2.2	-3.1		
		MCDB	-3.6	-1.4	4.2	11.1	17.9	22.7	25.0	24.0	20.4	12.1	3.9	-2.1		
	10%	WB	-6.7	-5.6	0.2	4.4	10.2	16.2	18.9	18.5	14.6	7.9	0.6	-5.0		
		MCDB	-6.1	-4.6	2.0	8.3	15.5	20.9	23.5	22.5	18.6	10.3	1.7	-4.2		
	5% DB	MDBR	10.0	12.2	13.0	12.2	12.6	12.1	12.2	12.1	11.8	9.2	8.5	9.2		
		MCDBR	10.6	12.0	14.8	16.6	19.9	16.2	14.7	14.7	15.5	13.6	9.8	9.1		
		MCWBR	9.6	10.5	10.8	10.4	11.0	8.1	7.5	7.9	9.0	8.9	7.4	8.1		
		MCWBR	10.4	11.7	13.3	16.0	16.8	13.7	13.0	12.9	14.1	12.1	9.0	9.1		
(35) Mean Daily Temperature Range	5% WB	MCWBR	9.5	10.2	9.7	10.4	10.0	7.7	7.5	7.6	8.7	8.4	6.9	8.2		
	taub		0.252	0.253	0.277	0.343	0.352	0.379	0.391	0.366	0.330	0.294	0.247	0.233		
		taud	2.221	2.283	2.347	2.374	2.433	2.389	2.388	2.464	2.546	2.577	2.401	2.255		
(42) Clear-Sky Solar Irradiance	Ebn at Noon		734	869	920	890	898	872	853	858	855	824	760	697		
		Edh at Noon	62	81	95	107	106	112	111	97	79	62	52	49		
	RadAvg		0.85	1.83	3.16	4.53	5.59	5.75	5.76	4.70	3.28	1.78	0.86	0.63		
(44) All-Sky Solar Radiation	RadStd		0.10	0.17	0.25	0.46	0.51	0.41	0.29	0.36	0.35	0.23	0.11	0.07		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	-0.67	-0.59	N/A	N/A	N/A
(47) Regional (7 neighbors)		N/A	N/A	+1.62	N/A	N/A	-0.37	N/A	N/A	N/A

Nomenclature: See separate page